

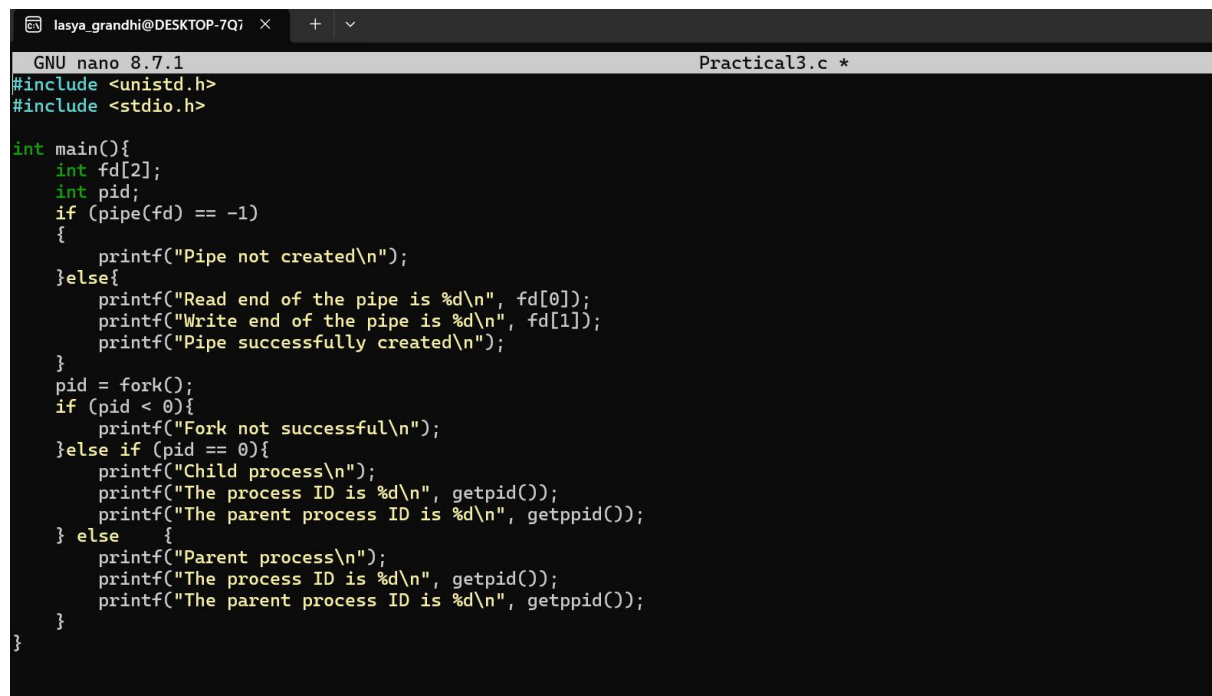
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3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

CODE:



```
GNU nano 8.7.1 Practical3.c *
#include <unistd.h>
#include <stdio.h>

int main(){
    int fd[2];
    int pid;
    if (pipe(fd) == -1)
    {
        printf("Pipe not created\n");
    }else{
        printf("Read end of the pipe is %d\n", fd[0]);
        printf("Write end of the pipe is %d\n", fd[1]);
        printf("Pipe successfully created\n");
    }
    pid = fork();
    if (pid < 0){
        printf("Fork not successful\n");
    }else if (pid == 0){
        printf("Child process\n");
        printf("The process ID is %d\n", getpid());
        printf("The parent process ID is %d\n", getppid());
    } else {
        printf("Parent process\n");
        printf("The process ID is %d\n", getpid());
        printf("The parent process ID is %d\n", getppid());
    }
}
```

EXECUTION :

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano Practical3.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc Practical3.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
Read end of the pipe is 3
Write end of the pipe is 4
Pipe successfully created
Parent process
The process ID is 871
The parent process ID is 807
Child process
The process ID is 872
The parent process ID is 802
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```

TASK2:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ps
  PID TTY          TIME CMD
  807 pts/2        00:00:00 bash
  904 pts/2        00:00:00 ps
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ top
top - 05:12:25 up 20 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15857.6 total, 15198.7 free, 612.8 used, 244.2 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 15244.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24984   15760  11672 S   0.0   0.1   0:00.57 systemd
    2 root        20   0    3180    2208   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    6 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   46 root        19  -1   50400   16860  15548 S   0.0   0.1   0:00.16 systemd-journal
   78 systemd+    20   0   22416   14536  12040 S   0.0   0.1   0:00.06 systemd-resolve
   85 root        20   0   35412   12364   9224 S   0.0   0.1   0:00.43 systemd-udev
  136 root        20   0    2888    2016   1884 S   0.0   0.0   0:00.03 chronyd-starter
  137 root        20   0    4472    3228   2984 S   0.0   0.0   0:00.01 cron
  138 message+    20   0    8820    5540   4824 S   0.0   0.0   0:00.06 dbus-daemon
  149 root        20   0   44096   29840  14696 S   0.0   0.2   0:00.21 networkd-dispat
  156 root        20   0   18436   9496   8252 S   0.0   0.1   0:00.08 systemd-logind
  179 syslog      20   0  220548    5808   4504 S   0.0   0.0   0:00.06 rsyslogd
  224 _chrony      20   0   23988   11580   7284 S   0.0   0.1   0:00.07 chronyd
  239 _chrony      20   0   12096    2500   1856 S   0.0   0.0   0:00.00 chronyd
  255 root        20   0  123460  32764  18156 S   0.0   0.2   0:00.16 unattended-upgr
  270 root        20   0    5492    2804   2616 S   0.0   0.0   0:00.00 agetty
  289 root        20   0    8648    5604   4760 S   0.0   0.0   0:00.01 login
  358 lasya_g+    20   0   22748   13512  11052 S   0.0   0.1   0:00.12 systemd
  362 lasya_g+    20   0   22920    4092   2088 S   0.0   0.0   0:00.00 (sd-pam)
  390 lasya_g+    20   0    6136    5472   3920 S   0.0   0.0   0:00.02 bash
  651 polkitd     20   0 306444    8452   7636 S   0.0   0.1   0:00.06 polkitd
  801 root        20   0    3184    1120    980 S   0.0   0.0   0:00.00 SessionLeader
  802 root        20   0    3200    1260   1108 S   0.0   0.0   0:00.07 Relay(807)
  807 lasya_g+    20   0    6268    5592   3916 S   0.0   0.0   0:00.05 bash
  919 lasya_g+    20   0    8156    6056   3964 R   0.0   0.0   0:00.00 top
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ /proc
-bash: /proc: Is a directory
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```